

LAYERSTONE

3D Printed Affordable Housing

Building homes layer by layer — making quality housing affordable for everyone

The Crisis

A housing emergency — not a market cycle

300K+

on London's social
housing waiting list

45-50%

of income spent
on rent in London

20,000

annual shortfall
in homes built

£550K+

average London
house price

Traditional construction is too slow, too expensive, and too labour-intensive.

The UK needs 300,000 homes per year — it builds ~230,000. London alone underbuilds by 20,000+ annually.

The problem isn't ambition or policy. It's capacity. We need a fundamentally different way to build.

The Solution

3D printed homes — faster, cheaper, better

40-60% Cheaper

£80K–£120K per unit
vs £200K–£300K
traditional

90% Faster

Structural printing
in 12–24 hours.
Total build 3–6 months.

80% Less Labour

3–5 workers on site
vs 15–25 for
traditional build

80-90% Less Waste

Precision printing
reduces material
waste dramatically

Layerstone = Housing Provider + 3D Printing Technology

We don't manufacture printers — we partner with Icon and COBOD, the global leaders in construction 3D printing. We provide what they lack: UK regulatory expertise, land access, tenant management, and social mission. Every home stays in our portfolio. Income-proportional rent. No profit-maximised pricing.

How It Works

From digital design to front door keys

1. Design

- Digital 3D model
- Unlimited customisation
- Architect approval

2. Print

- Robotic arm extrudes concrete layer by layer
- Walls printed in 12–24 hours
- 3–5 workers on site

3. Fit Out

- Roof, insulation, windows, doors
- Plumbing, electrics, heating
- Kitchen, bathroom, finishing

4. Move In

- Tenant selection (income-proportional)
- Building Control sign-off
- Keys handed over

Total timeline: 3–6 months from site preparation to move-in | Traditional: 12–18 months

The Business Model

Two viable paths — same mission

Path A: Community Interest Company → RHP

Factor	Path A
Setup time	2–4 weeks
Setup cost	~£10K–£17K
Grant access	Community Fund only (Y1–2)
Borrowing rate	Commercial (5–7%)
Board	Min 1 director
Y5 portfolio	65 units
Y5 asset value	£6.5M
Best for	Solo founder, proving model

Path B: Registered Housing Provider (Day 1)

Factor	Path B
Setup time	6–12 months
Setup cost	~£16K–£32K
Grant access	Affordable Homes Programme
Borrowing rate	Preferential (2–3%)
Board	5+ with expertise
Y5 portfolio	88 units
Y5 asset value	£8.8M
Best for	Team with housing expertise

Income-Proportional Rent (Both Paths)

Tenants pay 25–30% of gross income, capped at Local Housing Allowance rates.

£20K income → £417–£500/month | £30K → £625–£750 | £40K → £833–£1,000 | £50K+ → capped ~£1,100

A family earning £25K saves £1,000+/month vs London market rent.

Market Opportunity

A £12.8B market with a massive gap

TAM

£12.8B

UK affordable housing
spending annually

SAM

£4.2B

London + Birmingham
affordable housing

SOM

£2M–£5M

Layerstone Year 3
revenue target

Why Now

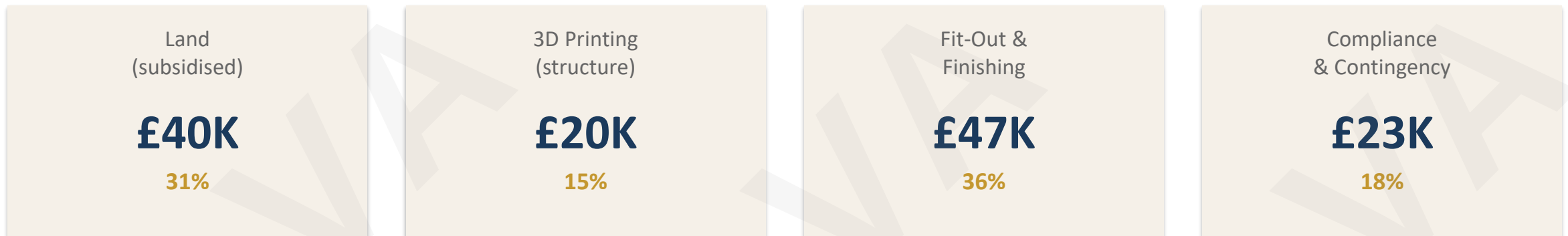
- £11.5B Affordable Homes Programme seeking partners
- Icon has proved 100+ homes in the US
- No UK company has commercialised 3DCP housing
- Construction labour shortage (25% deficit)
- Post-pandemic housing demand surge
- £163M Community Housing Fund available
- Brownfield Land Release Fund active
- Political urgency = council cooperation

Unit Economics

Every home costs less than half the traditional price

Unit Type	Layerstone Cost	Traditional Cost	Saving	Saving %
Studio	£71,300	£120,000	£48,700	41%
1-Bed	£98,325	£180,000	£81,675	45%
2-Bed House	£130,475	£250,000	£119,525	48%
3-Bed House	£162,725	£320,000	£157,275	49%

2-Bed Unit Breakdown (London)



Birmingham costs are 20–30% lower due to cheaper land. A 2-bed in Birmingham: ~£104K vs £250K traditional.

Financial Projections

Capital-intensive, asset-rich — every unit built is an investment

Path A (CIC)	Year 1	Year 2	Year 3	Year 4	Year 5
Units built	0	5	10	20	30
Total portfolio	0	5	15	35	65
Revenue	£0	£104K	£262K	£688K	£1.18M
Total costs	£17K	£644K	£1.28M	£2.53M	£3.81M
Asset value	£0	£500K	£1.5M	£3.5M	£6.5M
Net asset position	(£17K)	(£40K)	£480K	£1.65M	£3.86M

Path B (RHP) — Faster Growth, Higher Governance

Path B (RHP)	Year 1	Year 2	Year 3	Year 4	Year 5
Total portfolio	0	8	23	48	88
Revenue	£0	£406K	£928K	£1.68M	£2.81M
Asset value	£0	£800K	£2.3M	£4.8M	£8.8M
Net asset position	(£33K)	£359K	£1.65M	£3.85M	£7.44M

Housing is capital-intensive: negative cash flow reflects building investment, not losses. Every unit built is worth more than it cost.

Competitive Landscape

No UK competitor does what we do

Traditional Housing Associations

Organisation	Homes	New/Year	3D Printing?	Cost/Unit
Clarion	125K+	~2,500	No	£200K–£300K
L&Q	105K+	~3,000	No	£200K–£300K
Peabody	66K+	~2,500	No	£200K–£300K

Modular / Innovation Builders

Company	Technology	Status	Affordable Focus?	Cost/Unit
L&G Modular	Factory modular	Active	Mixed	£150K–£250K
ilke Homes	Volumetric modular	Administration (2023)	Yes	N/A — failed
TopHat	Steel-frame modular	Active	Yes	£120K–£200K

Layerstone's Unique Position

- Only UK company combining 3D printing + affordable housing social mission
- No factory overhead (ilke's fatal flaw) — printers deploy to site, keeping costs variable
- vs Housing Associations: 40–60% lower construction costs
- vs International 3D printers: UK regulatory expertise + land access + housing credentials
- Potential partners, not just competitors — large HAs need innovative builders

Social Impact

Phase 1: Affordable homes | Phase 2: Homeless rehousing & employment

Phase 1 (Years 1–3): Prove the Model

- Build 15–23 affordable homes (Path A/B)
- House families at 25–30% of income
- Save tenants £1,000+/month vs market rent
- Prove 3DCP works in UK regulations
- Create construction jobs
- Publish cost analysis and case studies
- Demonstrate model to councils and investors

Phase 2 (Year 3+): Social Programmes

- Allocate 10–20% units to homeless rehousing
- 3–6 months rent-free, then income-based
- 3D printer operator training for tenants
- Property maintenance training programme
- Community management roles for residents
- Earned rent reduction for programme workers
- Partnership with homelessness charities

Long-Term Targets

Metric	Year 5	Year 10
Homes delivered	65–88	300–500
People housed	150–250	800–1,500
Avg tenant saving vs market	40–60%	40–60%

Pilot Launch Strategy

London first, Birmingham second — prove before scaling

Months 1-3 Pre-Launch

- Legal setup (CIC or RHP) application
- Partner outreach (Icon, COBOD)
- Council engagement (Barking, Croydon, Enfield)
- Land search begins

Months 4-6 Site Secured

- Planning pre-application
- BBA certification started
- Structural testing
- Community consultation

Months 7-9 First Build

- Planning permission granted
- Printer deployed
- First 2-3 homes printed + fitted out
- Building Control sign-off

Months 10-18 Expand

- First tenants move in
- Case study published
- Remaining pilot units built
- Birmingham site assessment begins

Target boroughs: Barking & Dagenham, Croydon, Enfield — brownfield land, high housing need, council innovation appetite
Birmingham launch: Year 2 | Startup cost: £10K–£32K | Pilot build budget: £300K–£600K (grant-funded)

LAYERSTONE

**Building homes layer by layer.
Making quality housing affordable for everyone.**

65–88 homes by Year 5 | £6.5M–£8.8M in assets
40–60% cheaper than traditional | Income-proportional rent
London + Birmingham | Social mission at the core

Thank You
Questions & Discussion